



Phloem

The innermost layer of the tree's bark, called the phloem, transports food made by photosynthesis in the leaves.

Cambium

A thin layer of actively dividing cells, the cambium generates more xylem and phloem as the tree grows thicker.

Sapwood

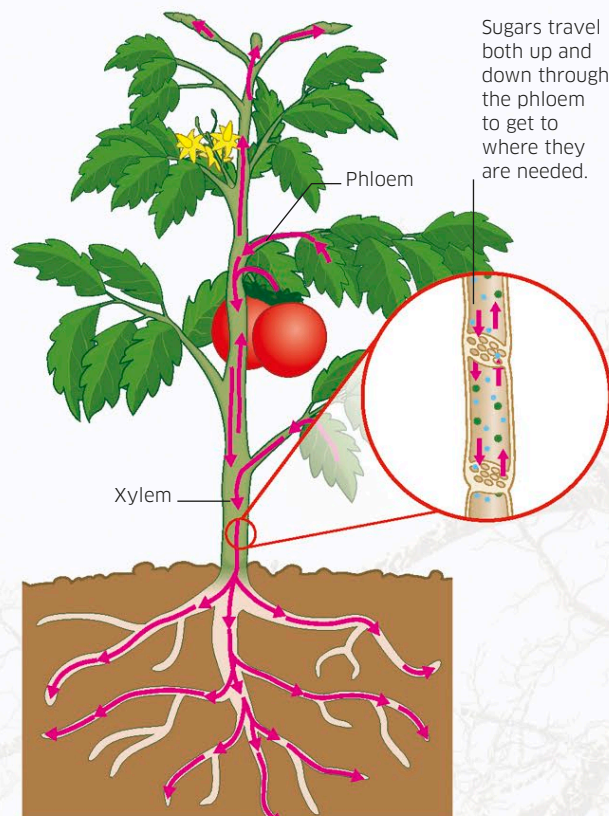
This contains the xylem vessels that stream water up the tree.

Heartwood

This is made of old xylem vessels that no longer carry water, but help support the weight of the tree.

Food distribution

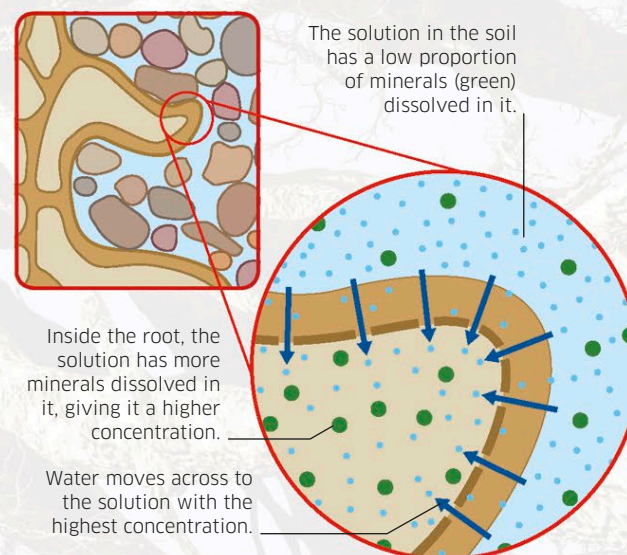
Sugars and other food are made in the leaves through photosynthesis (see pp.148-149). They are then carried through pipes called phloem—traveling to roots, flowers, and others parts that cannot make food for themselves.



Sugars travel both up and down through the phloem to get to where they are needed.

Osmosis

When cell membranes stretch between two solutions with different concentrations, water automatically passes across to the higher concentration by a process called osmosis. This happens in plant roots—where root cell membranes are situated between the weak mineral solutions found in soil and the higher concentrations inside the root cells.



The solution in the soil has a low proportion of minerals (green) dissolved in it.

Inside the root, the solution has more minerals dissolved in it, giving it a higher concentration.

Water moves across to the solution with the highest concentration.